

2 Junie / *June* 2004

Punte / Marks:50

Tyd / Time: 120 min

Punte : Intern <i>Marks : Internal</i>	Veranderinge <i>Changes</i>	Punte : Ekstern <i>Marks : External</i>
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WTTW 158 : CALCULUS

EKSAMEN / *EXAMINATION*Eksterne eksaminator / *External examiner*: Me/Ms R Möller

Van / Surname :

Voorname / *First names*:

Studentennummer / Student number:							—	Handtekening / Signature:
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Studierigting / <i>Field of Study</i> :	B	C	E	L	M	N	P	R	S	Z	Ander / Other
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Telefoonnummer gedurende examenperiode / <i>Telephone number during exam period</i>	
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Interne eksaminatore / Internal examiners	Interne eksaminatore / Internal examiners
Mev / Mrs A E du Preez	Dr S M Maepa
Prof JC Engelbrecht	Prof M J Schoeman

Lees eers die volgende instruksies

1. Slegs nie-programmeerbare sakrekenaars mag gebruik word.
2. Die vraestel bestaan uit bladsye 1 tot 10 (vrae 1-28) asook 'n merkleesvorm. Kontroleer of u vraestel volledig is.
3. Doen alle krapwerk op die blanko bladsy(e). Dit word nie nagesien nie.
4. As u meer as die beskikbare ruimte vir 'n antwoord nodig het, gebruik dan ook die blanko bladsy(e) en dui dit duidelik aan deur 'n **raam daarom te trek**.
5. Geen potloodwerk word nagesien nie.
6. Korreksievloestof (bv. Tipp-Ex) mag nie gebruik word nie.
7. Geen antwoordboek mag uit die lokaal geneem word nie
8. **Toon alle bewerkings duidelik**

First read the following instructions

1. Only non-programmable calculators may be used.
2. The question paper consists of pages 1 to 10 (questions 1-28) and an optic reader form. Check if your paper is complete.
3. Scribbling can be done on the empty page(s). This will not be marked.
4. If you need more than the available space for an answer, you may also use the empty page(s). Indicate it clearly by **framing** it.
5. Work in pencil will not be marked.
6. Use of correction fluid (e.g. Tipp-Ex) is not allowed.
7. No test scripts may be removed from the venues
9. **Show all computations clearly**

Beantwoord vrae 1 tot 22 op die MERKLESERVORM se KANT 1

Answer questions 1 to 22 on the OPTIC READER FORM on SIDE 1

Vraag 1 / Question 1Die waardeversameling van $f(x) = \sin^{-1}(3x + 1)$ isThe range of $f(x) = \sin^{-1}(3x + 1)$ is(Notasie / Notation: $\sin^{-1}x = \arcsin x = \text{bgsin}x$)

(1 a) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	(1 b) $\left[-\frac{\pi}{6}, \frac{\pi}{6}\right]$	(1 c) $\left[-\frac{2}{3}, 0\right]$	(1 d) $[-1, 1]$
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Vraag 2 / Question 2Bepaal / Find: $\lim_{x \rightarrow 1^+} \frac{x^2 - 3x + 2}{x^2 - 2x + 1}$

(2 a) $\lim_{x \rightarrow 1^+} \frac{2x-3}{2x-2} = \lim_{x \rightarrow 1^+} \frac{2}{2} = 1$	(2 b) $\lim_{x \rightarrow 1^+} \frac{1 - \frac{3}{x} + \frac{2}{x^2}}{1 - \frac{2}{x} + \frac{1}{x^2}} = 1$	(2 c) ∞	(2 d) $-\infty$
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Vraag 3 / Question 3 $\lim_{x \rightarrow \infty} \ln x^e =$ $x \rightarrow \infty$

(3 a) ∞	(3 b) e	(3 c) $-\infty$	(3 d) geen van hierdie nie / none of these
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Vraag 4 / Question 4As / If $f(x) = 3^{-x}$ dan / then $f'(x) =$

(4 a) 3^{-x}	(4 b) $\frac{1}{3^x \ln 3}$	(4 c) $-x 3^{-x-1}$	(4 d) $(-\ln 3) 3^{-x}$
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Vraag 5 / Question 5 $D_x(\cos^2 \frac{1}{x}) =$

(5 a) $(-\sin^2 \frac{1}{x}) (\frac{-1}{x^2})$	(5 b) $2 (\cos \frac{1}{x}) (-\sin \frac{1}{x})$
(5 c) $2 (\cos \frac{1}{x}) (\sin \frac{1}{x}) (\frac{1}{x^2})$	(5 d) geen van hierdie nie / none of these

Vraag 6 / Question 6 $D_x(\tan^{-1} e^{2x}) =$

(6 a) $\frac{e^{2x}}{1+e^{4x}}$	(6 b) $\frac{2e^{2x}}{1+e^{4x}}$	(6 c) $D_x(\tan^{-1} e^{2x}) = D_x \cot e^{2x} = -2e^{2x} \operatorname{cosec}^2 e^{2x}$
(6 d) $-2e^{2x} (\tan^{-2} e^{2x}) (\sec^2 e^{2x})$		

(Notasie / Notation: $\tan^{-1}x = \arctan x = \text{bgtan}x$)

Vraag 7 / Question 7

$$\int e^y (1 - e^{-y}) dy =$$

(7 a) $e^y \frac{(1-e^{-y})^2}{2} + C$	(7 b) $e^y - 1 + C$	(7 c) $e^y - y + C$
(7 d) geen van hierdie nie / none of these		

Vraag 8 / Question 8

$$\int \left(\frac{1}{t^2} - \sec^2 t \right) dt =$$

(8 a) $\ln(t^2) + \tan t + C$	(8 b) $\frac{-1}{t} - \tan t + C$	(8 c) $\frac{-2}{t^3} - 2\sec^2 t \tan t + C$
(8 d) geen van hierdie nie / none of these		

Vraag 9 / Question 9

Watter van die volgende is korrek? / Which of the following is / are correct?

(i) $\int (\sin x \cos x) dx = \frac{\sin^2 x}{2} + C$

(ii) $\int (\sin x \cos x) dx = \frac{-\cos^2 x}{2} + C$

(iii) $\int (\sin x \cos x) dx = \frac{-1}{2} \cos 2x + C$

(9 a) slegs (i) is korrek / only (i) is correct
(9 b) (i) en (ii) is korrek / (i) and (ii) are correct
(9 c) al die antwoorde is korrek / all the answers are correct
(9 d) geen van hierdie nie / none of these

Vraag 10 / Question 10

$$\int \frac{2u+1}{u} du =$$

(10 a) $2u + \ln u + C$	(10 b) $(u^2 + u) \ln u + C$	(10 c) $\frac{(2u+1)^2}{2} + C$
(10 d) geen van hierdie nie / none of these		

Vraag 11 / Question 11

$$\int \frac{(\ln y + 4)^3}{y} dy =$$

(11 a) $\frac{\mathcal{L}(\ln y + 4)^4}{2} + C$	(11 b) $\frac{(\ln y + 4)^4}{2} + C$	(11 c) $\ln y \frac{(\ln y + 4)^4}{2} + C$
(11 d) geen van hierdie nie / none of these		

Vraag 12 / Question 12

$$\int \sin^2\left(\frac{x}{2}\right) dx =$$

(12 a) $-\frac{1}{6}\cos^3\frac{x}{2} + C$	(12 b) $\frac{1}{2}[x - \sin x] + C$
(12 c) $\frac{1}{2}[x + \sin x] + C$	(12 d) geen van hierdie nie / none of these

Vraag 13 / Question 13

Laat / Let $f(x) = \cosh x$ en / and $g(x) = \sinh x$.

Vir watter waarde(s) van x sal f en g sny?

For which value(s) of x will f and g intersect?

(13 a) $x \approx 1.5$	(13 b) f en g sny oneindig veel kere / f and g intersect infinitely many times
(13 c) f en g sny mekaar nooit / f and g never intersect	
(13 d) geen van hierdie nie / none of these	

Vraag 14 / Question 14

Bepaal $\frac{dy}{dx}$ as $\cos(xy) = y$. / Find $\frac{dy}{dx}$ if $\cos(xy) = y$.

(14 a) $-\cos(xy)$	(14 b) $\frac{-y \sin(xy)}{1+x \sin(xy)}$	(14 c) $\frac{y \sin(xy)}{1-x \sin(xy)}$
(14 d) geen van hierdie nie / none of these		

Vraag 15 / Question 15

Vind die funksie f wat aan die volgende voorwaardes voldoen:

Find the function f that satisfies the following conditions:

$$f''(\theta) = e^{2\theta}, \quad f'(0) = 2, \quad f(0) = \frac{1}{4}$$

(15 a) $\frac{1}{2}e^{2\theta} + 2\theta + \frac{1}{4}$	(15 b) $\frac{1}{4}e^{2\theta} + \frac{3}{2}\theta + \frac{1}{4}$	(15 c) $\frac{1}{4}e^{2\theta} + \frac{3}{2}\theta$
(15 d) nie een van hierdie nie / none of these		

Vraag 16 / Question 16

Laat $\bar{\mathbf{a}} = \langle 2, 1, -1 \rangle$ en $\bar{\mathbf{b}} = \langle 0, 1, 2 \rangle$ vektore wees, dan is $\bar{\mathbf{a}} \times \bar{\mathbf{b}} =$

Let $\bar{\mathbf{a}} = \langle 2, 1, -1 \rangle$ and $\bar{\mathbf{b}} = \langle 0, 1, 2 \rangle$ be vectors then $\bar{\mathbf{a}} \times \bar{\mathbf{b}} =$

(16 a) $\langle 3, 4, 2 \rangle$	(16 b) -1	(16 c) $2\bar{\mathbf{i}} + 2\bar{\mathbf{j}} + \bar{\mathbf{k}}$	(16 d) $\langle 3, -4, 2 \rangle$
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Vraag 17 / Question 17

Die afstand vanaf die punt $P(-1, 2, 5)$ na die YZ – vlak is

The distance from the point $P(-1, 2, 5)$ to the YZ – plane is

(17 a) 1	(17 b) -1	(17 c) $\sqrt{30}$	(17 d) $\sqrt{29}$
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Vraag 18 / Question 18

'n Lyn L_1 het vergelykings $x = 3$, $\frac{y+1}{-2} = \frac{z-1}{3}$. 'n Punt op die lyn L_1 is

A line L_1 has equations $x = 3$, $\frac{y+1}{-2} = \frac{z-1}{3}$. A point on the line L_1 is

(18 a) $(0, -2, 3)$	(18 b) $(3, 1, 1)$	(18 c) $(3, -1, 1)$	(18 d) $(0, -1, 1)$
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Vraag 19 / Question 19

'n Lyn L_2 het vergelykings $\frac{1-x}{2} = \frac{y+1}{-1} = \frac{z}{4}$. 'n Vektor ewewydig aan die lyn L_2 is

A line L_2 has equations $\frac{1-x}{2} = \frac{y+1}{-1} = \frac{z}{4}$. A vector parallel to the line L_2 is

(19 a) $\langle 2, -1, 4 \rangle$	(19 b) $\langle -2, 1, -4 \rangle$	(19 c) $\langle -2, -1, 4 \rangle$	(19 d) $\langle 1, -1, 0 \rangle$
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Vraag 20 / Question 20

Laat f 'n funksie wees waarvan ons die volgende weet. / Let f be a function with the properties

$\lim_{x \rightarrow -\infty} f(x) = \infty$, $\lim_{x \rightarrow \infty} f(x) = \infty$, $\lim_{x \rightarrow 1^-} f(x) = \infty$ en / and $\lim_{x \rightarrow 1^+} f(x) = 3$.

Watter van die volgende is korrek? / Which of the following is / are correct?

(20 a) f het 'n horisontale asimptoot / f has a horizontal asymptote
(20 b) f het 'n vertikale asimptoot $x = 1$ / f has a vertical asymptote $x = 1$
(20 c) f het 'n vertikale asimptoot $y = 3$ / f has a vertical asymptote $y = 3$
(20 d) Niks kan van f gesê word nie / Nothing can be said of f

Vraag 21 & 22 / Question 21 & 22

Laat / Let $f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ a, & x = 0 \end{cases}$.

$\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} =$

(21 a) ∞	(21 b) 0	(21 c) die limiet bestaan nie / the limit does not exist	(21 d) 1
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f is kontinu op $\mathbb{R}$ as $a =$ / f is continuous on $\mathbb{R}$ if $a =$

(22 a) ∞	(22 b) 0	(22 c) a kan nie bereken word nie / a cannot be calculated	(22 d) 1
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Totaal / Total

[22]

BEANTWOORD ALLE VERDERE VRAE OP HIERDIE VRAESTEL
en TOON alle bewerkings duidelik aan
ANSWER ALL THE FOLLOWING QUESTIONS ON THE SCRIPT
and SHOW all computations clearly.

Vraag 23 / Question 23

Laat / Let $f(x) = e^{2x-x^2}$ met / with $f'(x) = (2 - 2x)(e^{2x-x^2})$

en / and $f''(x) = (2 - 8x + 4x^2)(e^{2x-x^2})$.

(23.1) Bepaal (indien moontlik) die lokale maksimumwaarde van f .

Find (if possible) the local maximum value of f .

[2]

(23.2) Vind die interval(le), indien enige, waarop f konkav na bo is.

Find the interval(s), if any, on which f is concave upward.

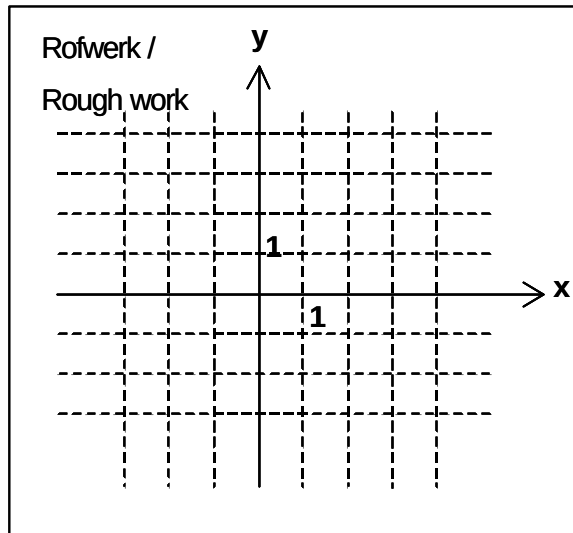
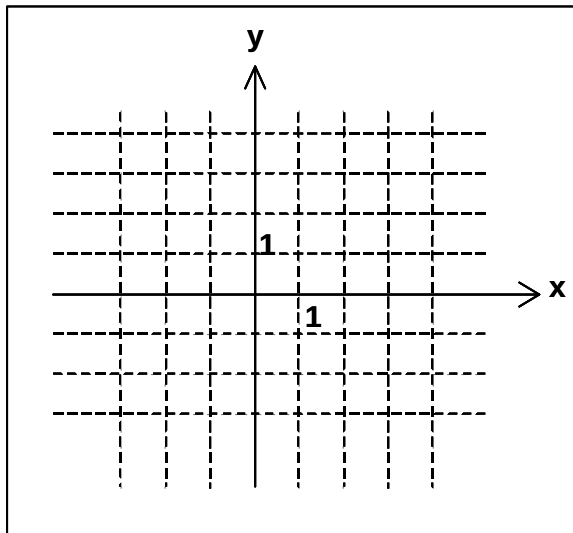
[3]

Vraag 24 / Question 24

Skets 'n funksie f wat kontinu is op $\mathbb{R}$ en die volgende voorwaardes bevredig.

Sketch a function f that is continuous on $\mathbb{R}$ and satisfies the following conditions.

- ♦ f is oneve / f is odd
- ♦ $f'(x) < 0$ as / if $0 < x < 2$
- ♦ $f'(x) > 0$ as / if $x > 2$
- ♦ $f''(x) > 0$ as / if $0 < x < 3$
- ♦ $f''(x) < 0$ as / if $x > 3$
- ♦ $\lim_{x \rightarrow \infty} f(x) = -2$



[4]

Vraag 25 / Question 25

Bepaal / Find

(25.1) $\int \frac{t+3}{t^2+1} dt.$

[3]

(25.2) $\int x\sqrt{2x+1} dx.$

[3]

Vraag 26 / Question 26

Laat / Let $y = (x + 1)^{\frac{3}{x}}$.

(26.1) Bepaal / Find $\frac{dy}{dx}$.

[3]

(26.2) Bepaal / Find $\lim_{x \rightarrow \infty} y$.

[3]

Vraag 27 / Question 27

Stel vas of die lyne L_1 en L_2 mekaar sny of nie. Indien L_1 en L_2 mekaar sny, gee ook die koördinate van die snypunt.

Determine whether the lines L_1 and L_2 intersect or not. If L_1 and L_2 intersect, give the co-ordinates of the point of intersection.

$$L_1 : \begin{cases} x = 4 + t \\ y = 2 \\ z = 3 + 2t \end{cases}, t \in \mathbb{R} \quad L_2 : \begin{cases} x = 2 + 2s \\ y = 2s \\ z = -1 + 4s \end{cases}, s \in \mathbb{R}$$

[4]

Vraag 28 / Question 28

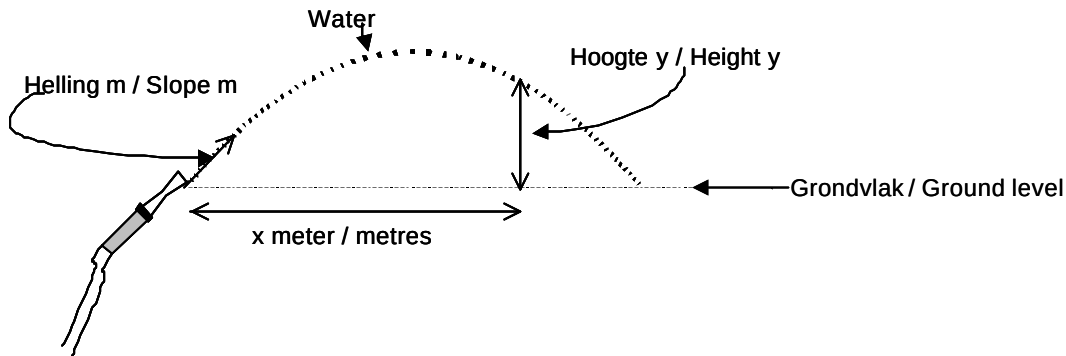
Dit kan aangetoon word dat die straal water wat uit 'n brandslang spuit, 'n hoogte van $y = -\frac{1}{80}(1 + m^2)x^2 + mx$ bokant 'n punt x meter vanaf die spuitkop bereik.

m is die helling van die spuitkop en neem aan dat die spuitkop op grondvlak gehou word.

It can be shown that the stream of water emitted by a fire hose will have height

$y = -\frac{1}{80}(1 + m^2)x^2 + mx$ *above a point located x metres from the nozzle.*

m *is the slope of the nozzle and the nozzle is assumed to be on ground level.*



(28.1) As $m = 2$, bepaal die afstand x waar 'n maksimum hoogte bereik sal word.

If $m = 2$, determine the distance x that will result in maximum height

[2]

(28.2) As 'n brandweerman $x = 40$ meter weg van 'n gebou moet staan, teen watter helling moet hy die spuitkop hou om die maksimum hoogte te bereik.

If a firefighter must stand $x = 40$ metres away from a building, at what slope must he aim to reach the maximum height.

[2]